Codimension-4 degenerations of nilpotent orbit closures and the symmetric pair $(\mathfrak{gl}_{p+q}, \mathfrak{gl}_p + \mathfrak{gl}_q)$

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1. Review of Kraft–Procesi's minimal degenerations
2. Codimension-4 degenerations
3. Classification of the fixed point loci $S^{\theta}_{\lambda,\mu}$

Minimal degenerations

Nilpotent $\mathrm{SL}_n(\mathbb{C})$-orbits in $\mathfrak{sl}_n(\mathbb{C})$ are indexed by partitions $\lambda \vdash n$.

$$\mu < \lambda \quad \Longleftrightarrow \quad \mathbb{O}_\mu \subset \overline{\mathbb{O}}_\lambda \setminus \mathbb{O}_\lambda.$$

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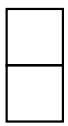
Hasse diagrams in $\mathfrak{sl}_2$ and $\mathfrak{sl}_3$

$\mathfrak{sl}_2$



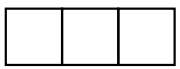
(2)

$xy - z^2 = 0 \mid A_1 = a_1$



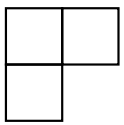
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$\mathfrak{sl}_3$



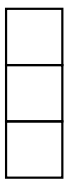
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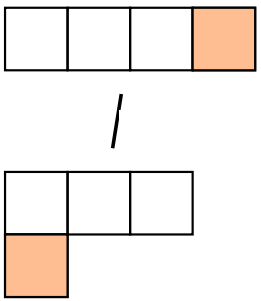
$(2,1)$

$\mid a_2$



$(1,1,1)$

A_r/a_r singularity and row/column removal



l

$(r+1)$

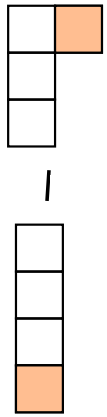
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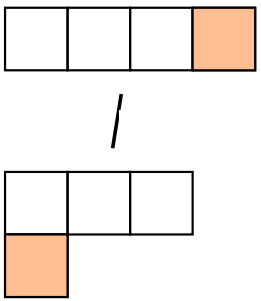
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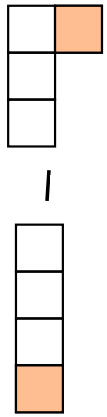
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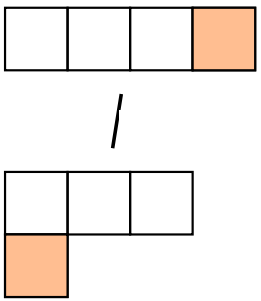
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Kleinian singularity $\rightsquigarrow$ dim 2

$a_r = \overline{\mathbb{O}}_{\min}(\mathfrak{sl}_{r+1}) \leftarrow T^*\mathbb{P}^r$

$\dim a_r = 2r$

A_r/a_r singularity and row/column removal



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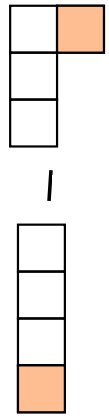
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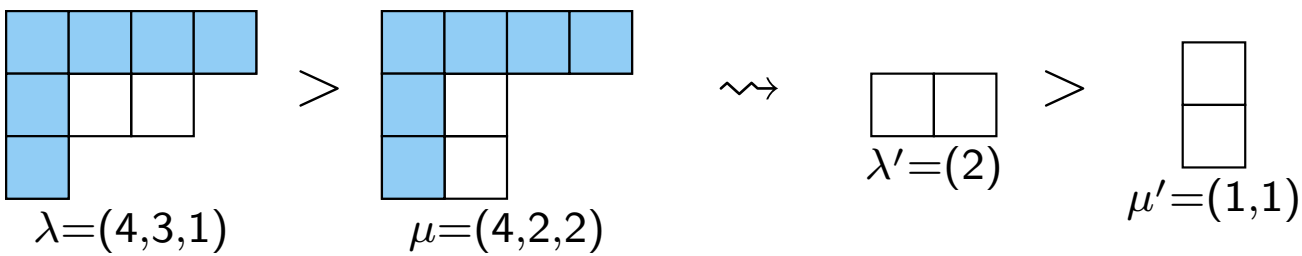
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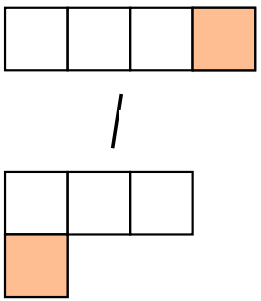
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Kraft–Procesi's removal of common rows and columns (blue boxes)



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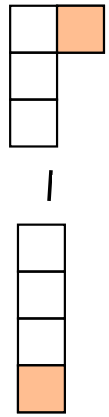
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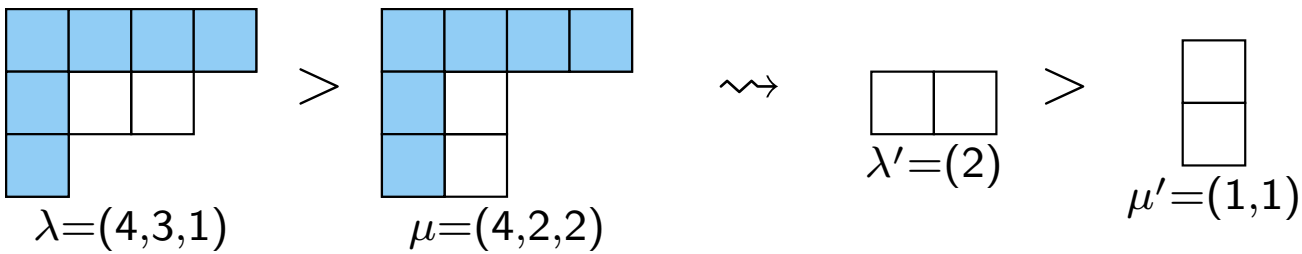
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Codim-4 degenerations: $a_2 = \overline{\mathbb{O}}_{\min}(\mathfrak{sl}_3)$ or two steps of A_r's.

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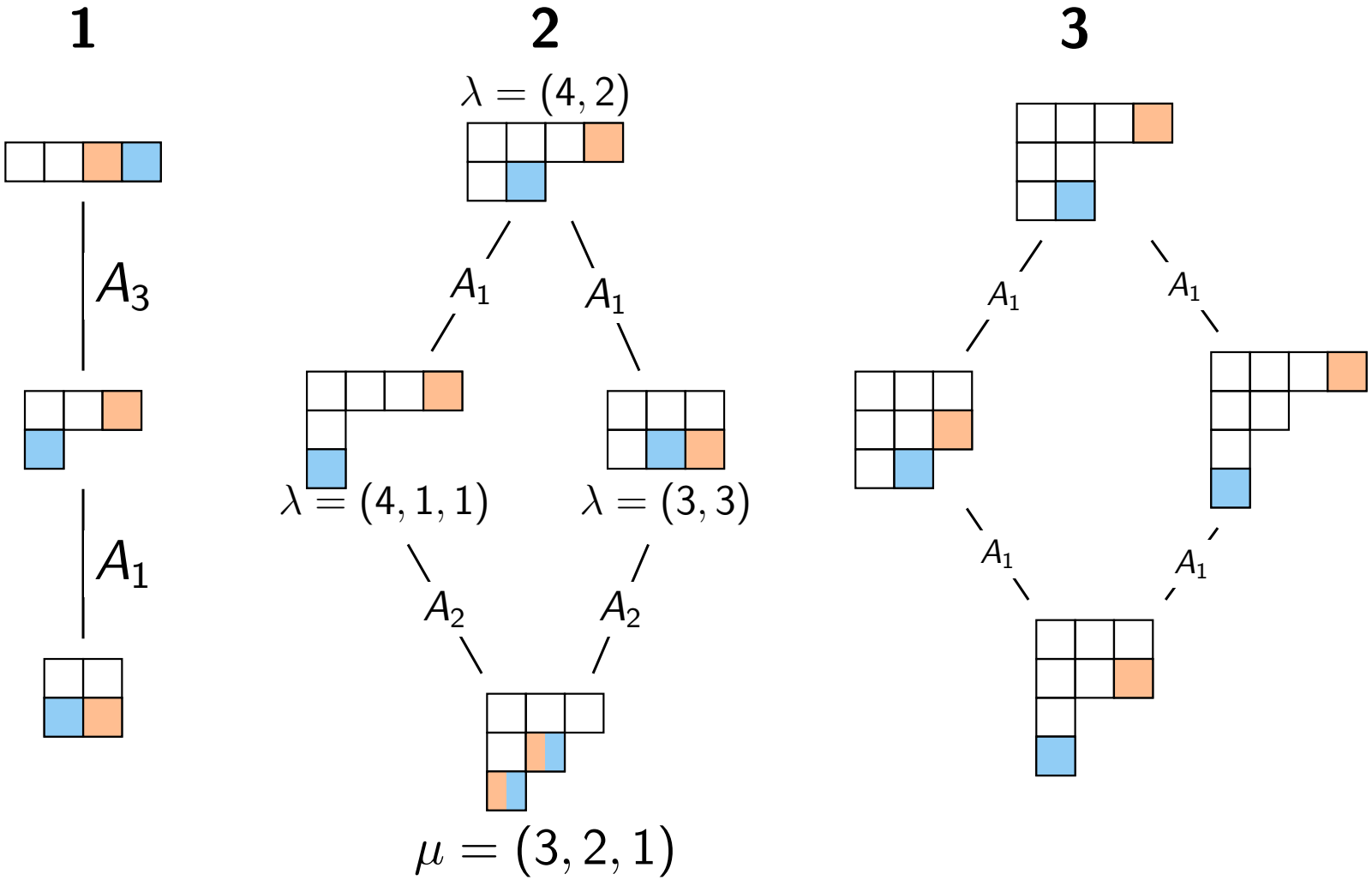
If the pair (μ', λ') is obtained from (μ, λ) through compatible signed row and column removals, then $S^{\theta}_{\lambda,\mu} \simeq S^{\theta'}_{\lambda',\mu'}$.

Codimension 4 degenerations

Examples where intermediate orbits occur.

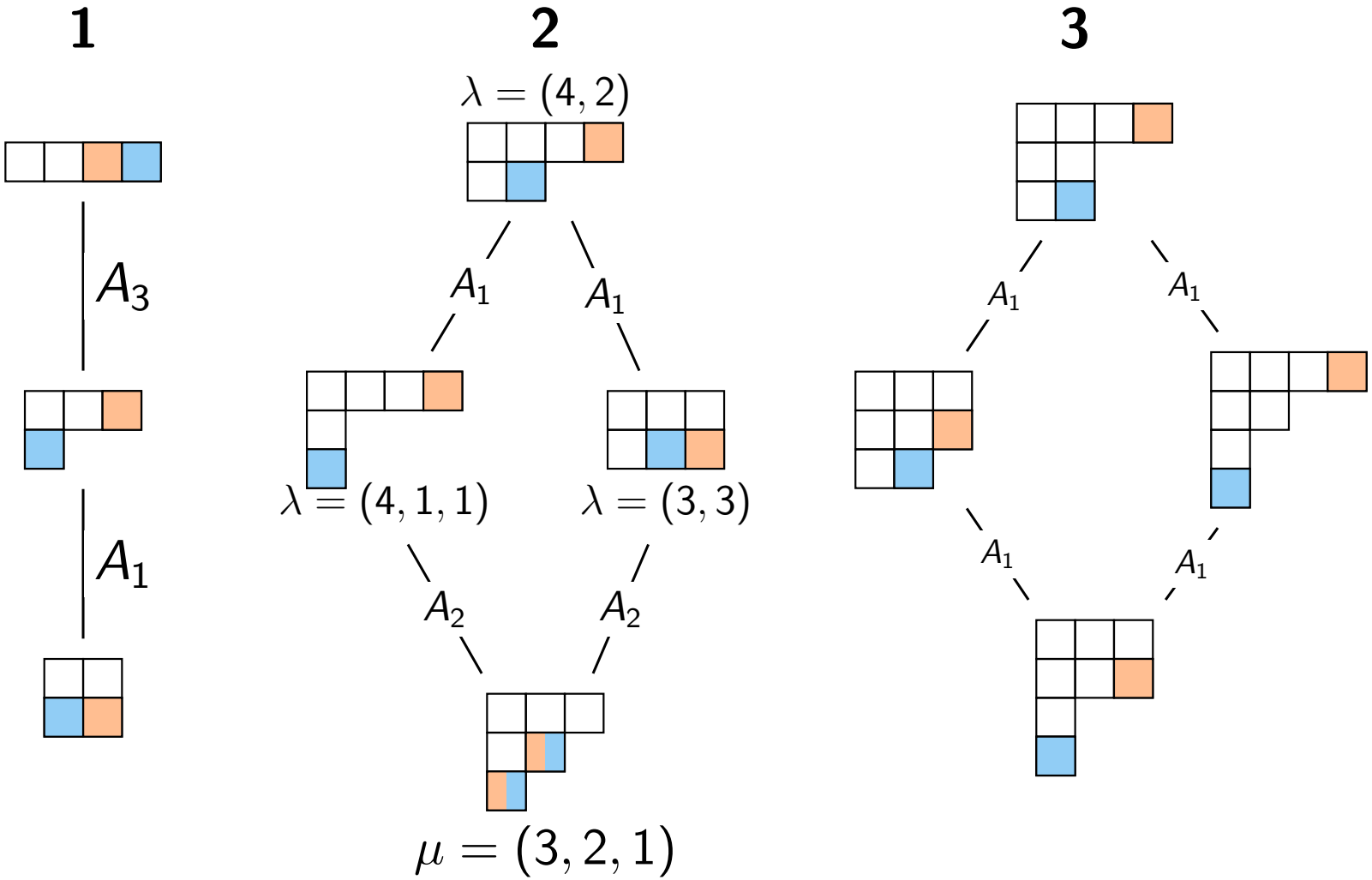
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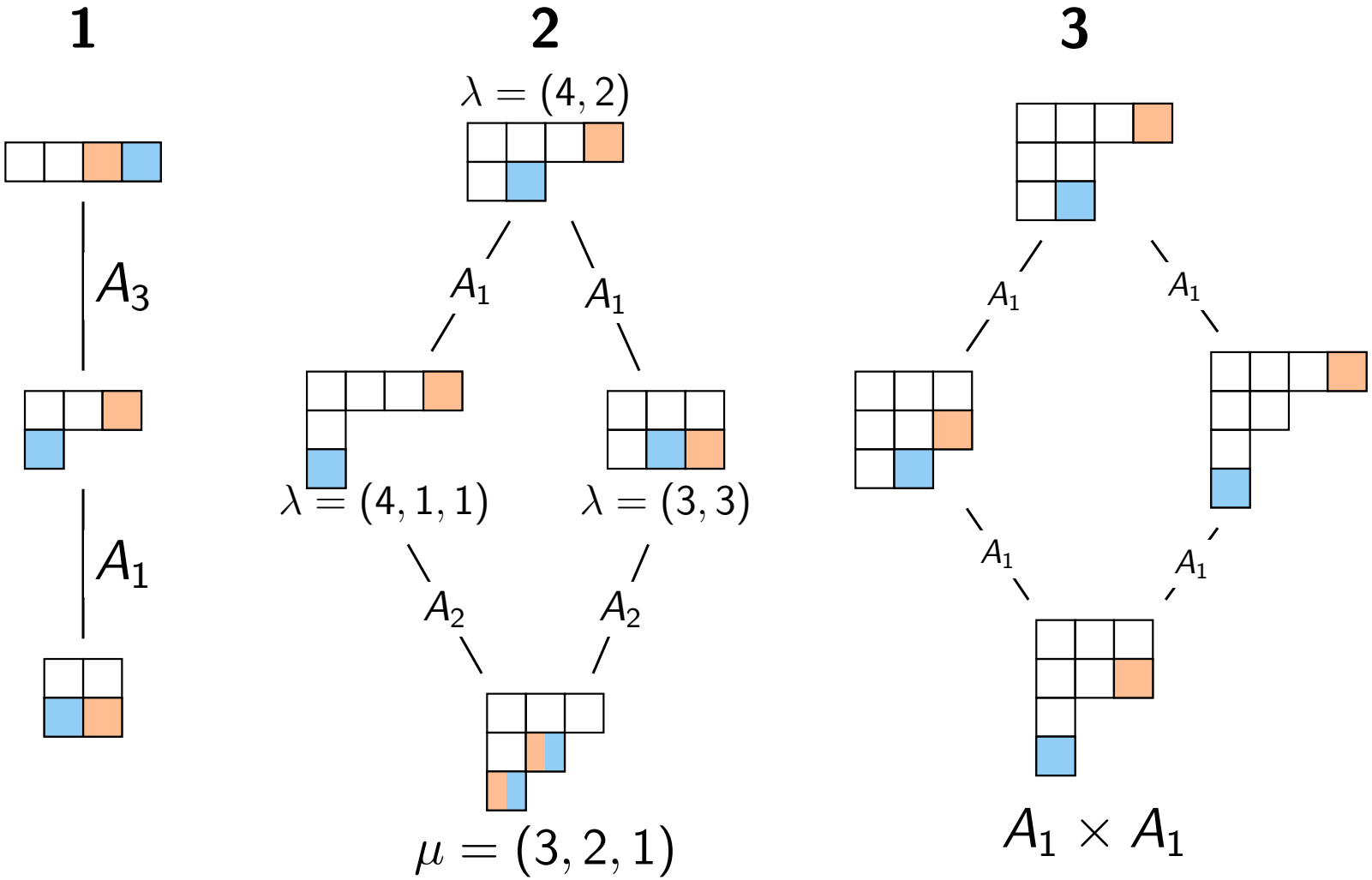
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Minimal models of codimension-4 degenerations

A combinatorial classification

A pair $\lambda > \mu$ with $\dim S_{\lambda,\mu} = 4$, after simultaneous row and column removals, falls within one of the following cases:

1. $(n) > (n-2, 2)$
2. $(m, n) > (m-1, n, 1)$
3. $(m, k, n) > (m-1, k+1, n-1, 1)$

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Remark:

- In Case 1, $\overline{\mathbb{O}}_{(n)} = \mathcal{N}$, the nilpotent cone. The fixed-point locus $S^{\theta}_{\lambda,\mu}$ is described in [Sekiguchi' 84]
- In Case 3, $S_{\lambda,\mu} \simeq A_{m-k-1} \times A_{n-1}$. The fixed point loci X^{θ}, for $X \simeq A_r$ Kleinian singularity, is described in [H. 26]
- Case 2 has not been studied in general. Note that Case 2 includes the minimal nilpotent orbit closure a_2 in $\mathfrak{sl}_3(\mathbb{C})$.

Ambient variety $S_{\lambda,\mu}$ in Case 2

Proposition 2 (H., in preparation)

For any pair $\lambda = (m, n) > \mu = (m-1, n, 1)$, we have

$$S_{\lambda,\mu} \simeq \operatorname{Spec} \mathbb{C}[x_1, x_2, x_3, y_1, y_2, y_3, z_1, z_2]/I_2(M),$$

where

$$M = \begin{pmatrix} z_1^n & x_1 & x_3 \\ y_1 & -(z_1+z_2) & x_2 \\ y_3 & y_2 & (-1)^{m-n-1} z_2^{m-n} \end{pmatrix},$$

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Examples of M:

$(4,2) > (3,2,1)$

$$\begin{pmatrix} z_1^2 & x_1 & x_3 \\ y_1 & -(z_1+z_2) & x_2 \\ y_3 & y_2 & -z_2^2 \end{pmatrix}$$

$\overline{\mathbb{O}}_{\min}(\mathfrak{sl}_3) = a_2$

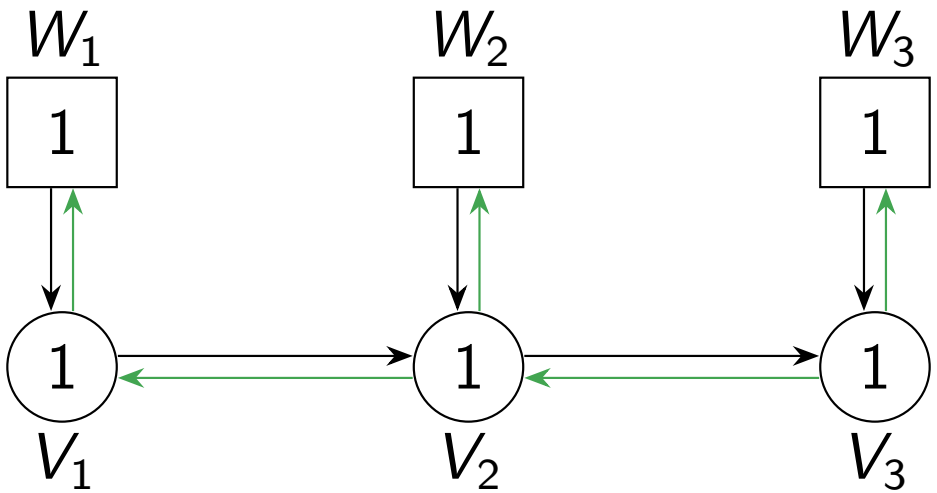
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Idea of proof: realize $S_{\lambda,\mu}$ as a quiver variety

[Maffei '05] The Nakajima–Maffei isomorphism gives

$$\Phi : \mathfrak{M}_0(\mathbf{v}, \mathbf{w}) \xrightarrow{\sim} S_{\lambda,\mu}.$$

For example, for $(4, 2) > (3, 2, 1)$, take $\mathbf{v} = \mathbf{w} = (1, 1, 1)$:

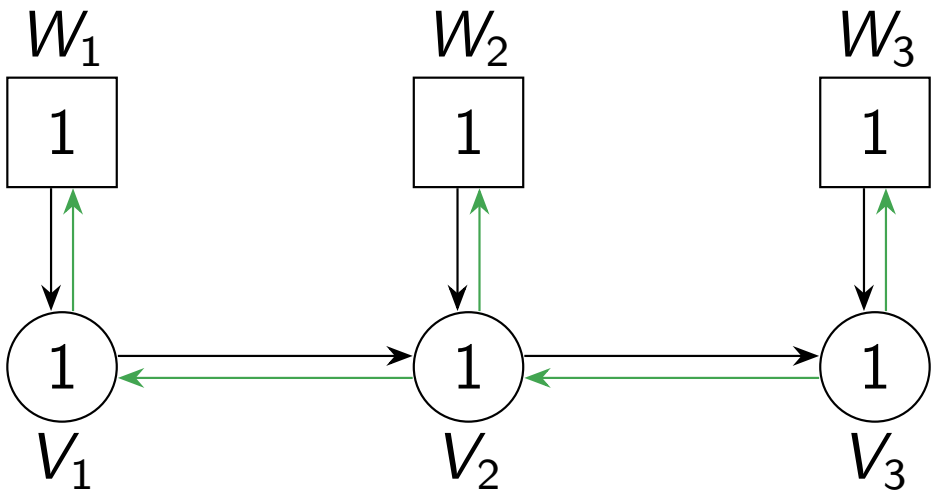


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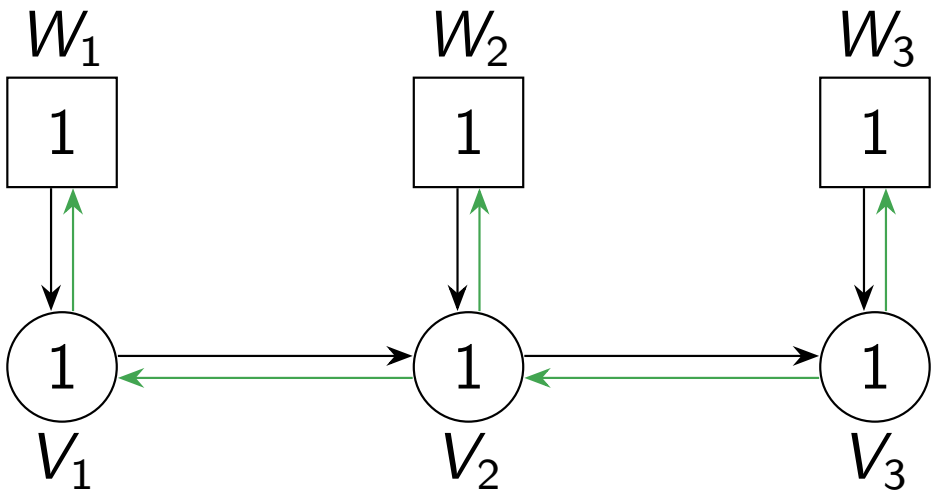
The generators x_i, y_i, z_i are given by paths between the framing spaces (squares). Relations come from associated moment maps.

Idea of proof: realize $S_{\lambda,\mu}$ as a quiver variety

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$z_1 : W_1 \to W_1, \qquad z_2 : W_3 \to W_3$

$x_1 : W_1 \to W_2, \qquad x_2 : W_2 \to W_3, \qquad x_3 : W_1 \to W_3$

Main result: description of the fixed point loci

Theorem (H., in preparation)

For $\lambda = (m, n) > \mu = (m-1, n, 1)$, the fixed-point locus $S^{\theta}_{\lambda,\mu}$ is isomorphic to one of the following varieties:

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1. $\operatorname{Spec} \mathbb{C}[x, y] = \mathbb{A}^2$
2. $\operatorname{Spec} \mathbb{C}[x, y, u, v]/((x, y)(u, v))$, two $\mathbb{A}^2$ meeting at a point.
3. $\operatorname{Spec} \mathbb{C}[x, y, z, u]/(xu, yu, zu, yz)$, two $\mathbb{A}^2$ meeting along a line, and an additional $\mathbb{A}^1$ meeting them at the origin.
4. $\operatorname{Spec} \mathbb{C}[x, y, z, u]/(xy, yz, zu)$, three $\mathbb{A}^2$ with two pairs meeting along lines.
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*Every irreducible component is $\mathbb{A}^2$, $\mathbb{A}^1$ or a point. Each irreducible component corresponds to a nilpotent K-orbit in $\mathfrak{p}$.

Idea of proof: involutions of quiver varieties

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Idea of proof: involutions of quiver varieties

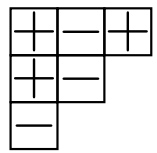
Recall the Nakajima–Maffei isomorphism $\Phi : \mathfrak{M}_0(\mathbf{v}, \mathbf{w}) \xrightarrow{\sim} S_{\lambda,\mu}$. We construct an involution θ_Q of the quiver variety $\mathfrak{M}_0(\mathbf{v}, \mathbf{w})$ such that $\Phi \circ \theta_Q = \theta \circ \Phi$. Then we have

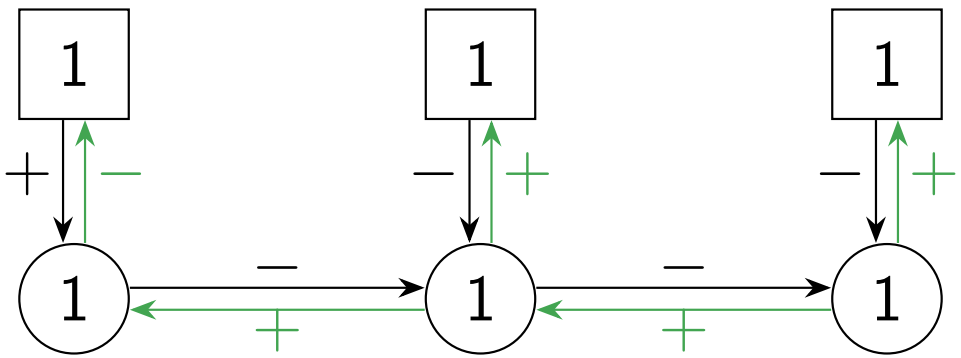
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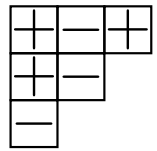
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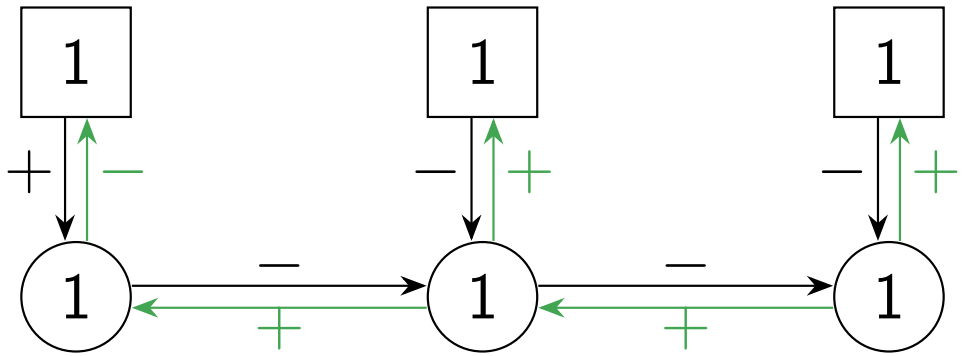


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We have $\theta(z_i) = -z_i,\ \theta(x_1) = -x_1,\ \theta(y_2) = -y_2$, resulting in that

$$S^{\theta}_{\lambda,\mu} \simeq \operatorname{Spec} \mathbb{C}[x_2, x_3, y_1, y_3]/(x_2 y_3,\ x_3 y_3,\ x_3 y_1).$$

Example: signed Hasse diagram for $\lambda = (4,2) > \mu = (3,2,1)$

$\mathfrak{g} = \mathfrak{sl}_6(\mathbb{C}),\ \sigma = \mathsf{Ad}(I_{3,3})$

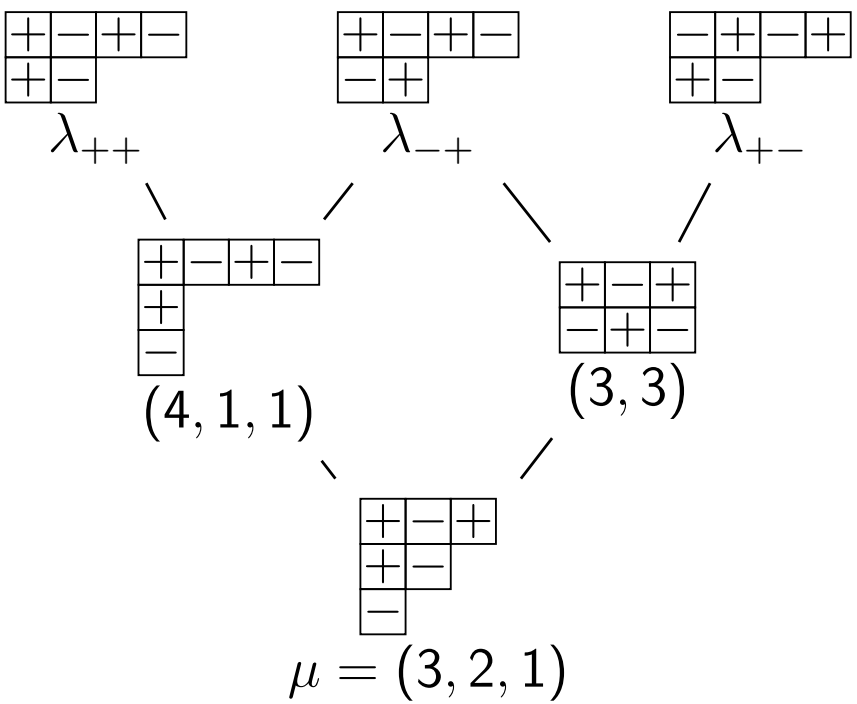
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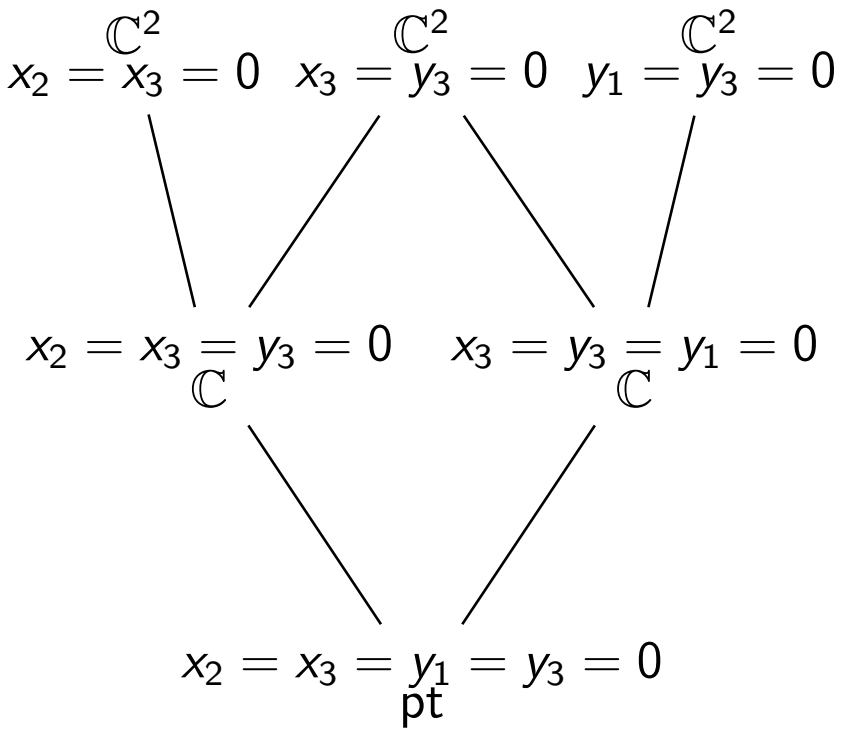
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$$S^{\theta}_{\lambda,\mu} \simeq \mathrm{Spec}\,\mathbb{C}[x_2, x_3, y_1, y_3]/(x_2y_3, x_3y_3, x_3y_1)$$





Thank you!